Kosuke Kikuchi

Nishinomiya, Hyogo, Japan
Software Engineer / Backend / Data Reliability Engineer
GitHub: https://github.com/kousukekikuchi1984

Summary

Software engineer specializing in **mission-critical data integrity**. I design systems where incorrect data must not become "confirmed truth," such as **accounting outputs and point-balance ledgers**. I prioritize **correctness, reproducibility, and explainability** under operational constraints (retries, partial failure, asynchronous boundaries), and I am known for making pragmatic engineering decisions—including when to **stop unsafe optimizations**. I currently work at **NTT-ME Corporation** on the **AI Property REsilience Business**, a business transferred from Tractable in the UK, where I am working to understand, stabilize, and improve the inherited system.

Core Strengths

- **Data integrity design:** idempotency, state transitions, failure modes, re-execution strategy, auditability
- **Production troubleshooting:** root-cause analysis (DB deadlocks, memory leaks), prevention by structural fixes
- **Backend architecture:** long-term maintainability, clear boundaries of responsibility, operationally sound workflows
- **Cross-functional delivery:** align stakeholders by clarifying constraints, risks, and alternatives (servant-style leadership)

Technical Skills

- **Backend:** Python (primary), Rust / Go / Java (limited)
- **Frontend:** TypeScript (production), Flutter (limited)
- **Databases/Data:** MySQL (primary), DuckDB, AWS Athena
- **Cloud/Infra:** AWS (ECS, EKS, Lambda, CloudFront, CloudWatch, EventBridge, Step Functions)
- **Data tools:** Digdag (code-level analysis), Embulk (code-level analysis)
- **Other:** HULFT (file transfer integration), Linux (conceptual familiarity)

Experience

NTT-ME Corporation — Software Engineer

Jun 2026 – Present

Department: Social Infrastructure Design Department, AI Property REsilience Business

Scope: Software engineering for the AI Property REsilience Business, transferred from Tractable in the UK. Currently focused on understanding the inherited system, operational model, and technical assumptions after the business transfer.

Current focus

- Understand the existing architecture, codebase, data flows, and operational workflows.
- Identify technical risks, knowledge concentration, and improvement opportunities required for stable post-transfer operations.

- Organize business, operational, and technical assumptions into practical engineering context for future development and maintenance.

YUZURIHA — Engineer (Backend / Platform)

Jan 2021 – May 2026

Scope: Technical direction for a ~10-engineer product team; design/code reviews as a quality gate; sub-project leadership (3–4 people); first-round engineering interviews.

Selected accomplishments

- **Prevented accounting CSV misstatement before launch** by identifying a latent edge case and correcting the design to match accounting rules (avoided costly post-release reconciliation).
- **Built safe large-scale ingestion for credit-card data (300K records, 1-hour SLA)** by redefining inconsistent external rules, deriving a safe parallelization unit (user-level key), separating validation from apply, and enforcing idempotent re-execution; achieved ~6 minutes ingestion in practice.
- **Led high-risk cutover of file-transfer integration (HULFT / OS / crypto changes)** across 4 stakeholders; re-evaluated trade-offs when new constraints emerged and executed a one-shot cutover with rollback/monitoring plans—completed without critical incidents.
- **Stopped unsafe "performance optimization"** by proving the bottleneck was MySQL commit fsync (not app CPU), proposing alternatives (commit reduction, responsibility split), and preventing a design fork that risked data integrity.
- **Resolved production incidents structurally:** fixed a post-release **memory leak** caused by persistent caching of DB Engine objects; eliminated recurring **DB deadlocks** by standardizing lock acquisition order and codifying the rule.

Gunosy — Data Reliability Engineer

May 2019 – Dec 2020
Ensured correctness and explainability across large-scale data workflows spanning ingestion, aggregation, and business operations.

- Designed revenue data finalization workflows with strict "freeze" semantics and resolved non-obvious Digdag behavior via code-level analysis.
- Built workflow templates to democratize the platform and reduced operational noise by designing convergence-by-structure.
  Reference: https://tech.gunosy.io/entry/escape-from-pitfall

PERSOL CAREER — Systems Engineer

Oct 2017 – Apr 2019
Selected: on-prem Hadoop for ML systems; docs for ML project execution; engine maintenance & review.

IP ONWEB JAPAN — Data Mining Engineer

Feb 2016 – Sep 2017
Analytics/ops for advertising delivery and revenue data; anomaly detection from distribution shifts; root-cause analysis of "looks normal but wrong" data states.

Spotlight — Engineer

Apr 2014 – Jan 2016
O2O product development; improved report generation performance and operational efficiency.

Technical Writing & Contributions

- HTTP/3 (QUIC) partial Japanese translation: https://http3-explained.haxx.se/ja
- Cache locality and performance reasoning (Qiita): https://qiita.com/kikuchi-yzrh/items/56b2368e4473e19ec1d9
- SQLAlchemy IdentityMap deep-dive (Qiita): https://qiita.com/kikuchi-yzrh/items/9680fd85f0dcb474ef18
- MySQL full-text search edge case note (Qiita): https://qiita.com/kikuchi-yzrh/items/baaab16cc1258e81e476
- Digdag retry duplication pitfall (Gunosy Tech Blog): https://tech.gunosy.io/entry/escape-from-pitfall